

ML System Design -1

- > Lifecycle of ML System
- > Drift
- > ML flow example

Lifecycle of ML System

- 1> Process the data → Preprocessing → Cleaning, Null values, normalization, encoding
- 2> Model Building → Baseline version of model
→ Carry hyperparameter tuning
- 3> Productionize model → Take the best obtained model
- 4> Test the model → Pytest
- 5> Deployment → Flask App(local)
↳ GET, POST
↓
Model → I/O
- 6> CI/CD → Github Actions
AWS → ECR, ECS
↳ Clusters
↳ Task
- 7> Monitoring

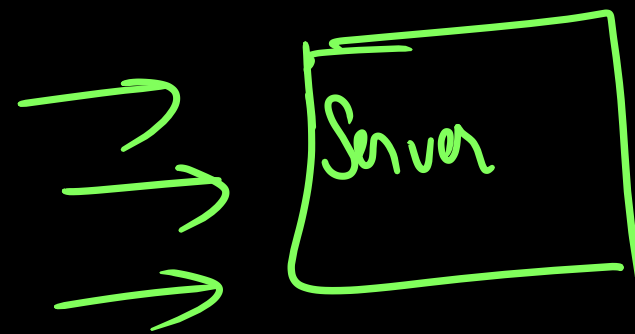
✓
{ ① Request per second (QPS)

{ ② Latency

③ Performance Metrics

↳ MSE, RMSE

↳ Accuracy, Precision, Recall



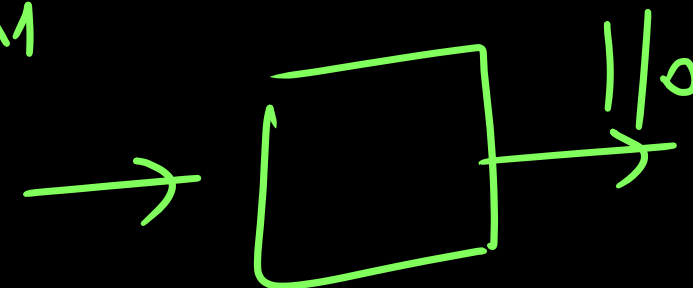
1000 QPS

5000 QPS

Drift



$M-6$ M



1> Concept Drift

2> Covariate Drift

✓
 $M-12$

✓
 $M-6$

M



{ ✓ $P_{\text{train}}(X, Y) = \text{Training data distribution}$
✓ $P_{\text{test}}(X, Y) = \text{Test data distribution}$

$P_{\text{train}} \neq P_{\text{test}} \rightarrow \text{Drift has occurred}$

Covariate Drift

↳ Input feature distribution changes (X)

$$\underbrace{P_{\text{train}}(X)} \neq \underbrace{P_{\text{test}}(X)}$$

$$\checkmark P_{\text{train}}(X, Y) \neq P_{\text{test}}(X, Y) \rightarrow \text{Drift}$$

$$\begin{array}{cc} \parallel & \checkmark \\ P_{\text{train}}(X) & P_{\text{train}}(Y/X) \end{array}$$

$$\begin{array}{cc} \parallel & \checkmark \\ P_{\text{test}}(X) & P_{\text{test}}(Y/X) \end{array}$$

House price prediction

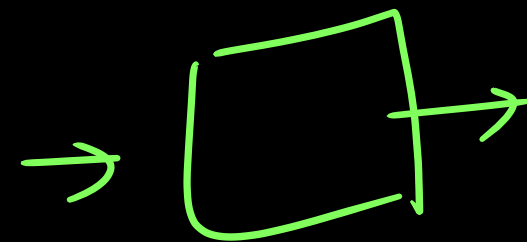
↳ Trained : Urban Area ✓

$P(X)$
train

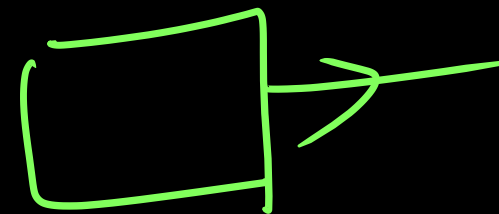
↳ Test : Rural Area

$P_{\text{test}}(x)$

High



Poor quality

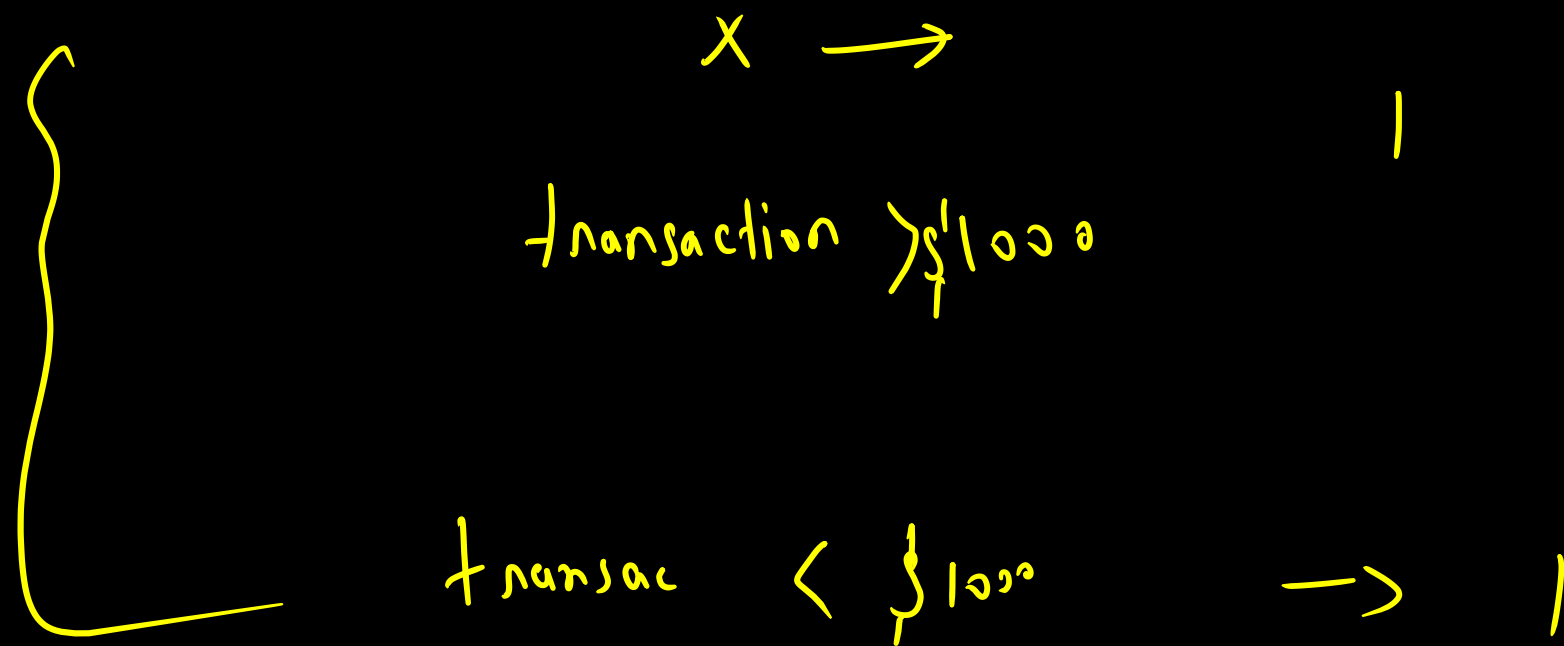


Concept drift

↳ Relation b/w Input features (X)
& output (y) changes.

$$\text{↳ } \underbrace{P(y/x)_{\text{train}}}_{\text{train}} \neq P(y/x)_{\text{test}}$$

Fraud detection

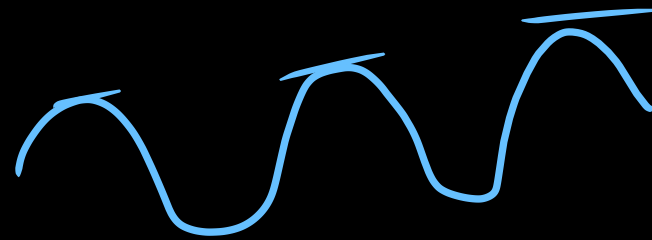


Concept drift

✓ 1> Sudden → House price change during COVID

✓ 2> Incremental →

3> Recurring →



✓ Method to detect Drift

① Visual Inspection ✓

② Statistical Test → KS test, Chi-square test ✓

③ Model performance test → MSE, RMSE,
Accuracy, Precision, Recall. ✓

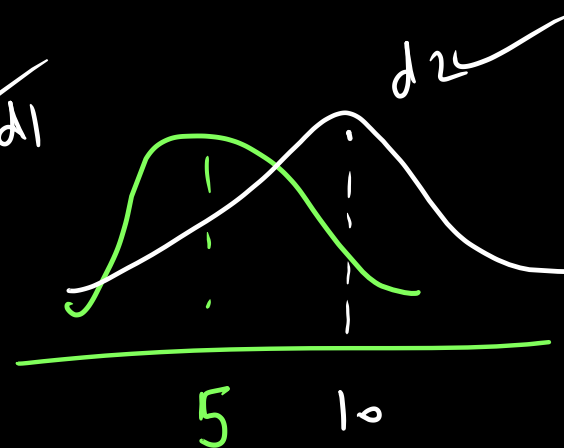
KS test (Kolmogorov - Smirnov test)

- ① Used for numerical variable ✓
- ② Compare two samples (dataset) are coming from same distribution

H_0 : Both samples come from same dist. $\rightarrow$ No drift

H_a : Both sample from diff. dist.

✓ $p\text{-value} < \text{Significance_level}$
"0.05"



$\rightarrow$ Reject H_0 (Accept H_a)

$\rightarrow$ Drift

Chi-square test-

① Used for categorical variable ✓

H_0 : No drift

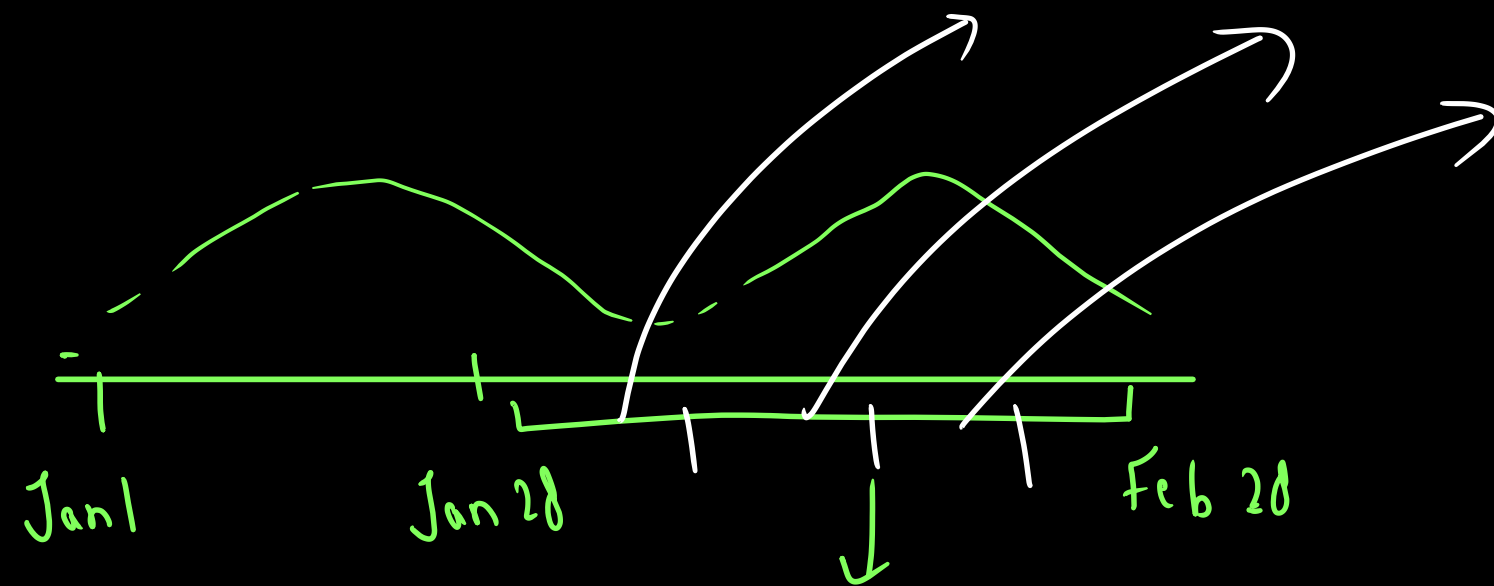
H_a : Drift

P-value < Significance level
||
0.05

↪ Reject H_0 (Accept H_a)

||
Drift

→ Break until 22:18 PM



↓

Data 1

↓

test data

↳ MSE

↳ MAE

↓

Ref-MSE

Ref-MAE

Data 2

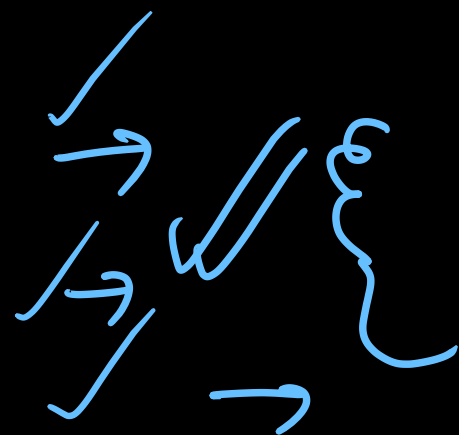
↓

↳ MSE

↳ MAE

$$MAE = \sum |y^{(i)} - \hat{y}^{(i)}|$$

$$MSE = \sum_{i=1}^n (y^{(i)} - \hat{y}^{(i)})^2$$



					Metrics	
☐	Run Name	Created	Duration		MAE	MSE
☐	run_2011-02-15 00:00:00 : 2011-02-21 23:00:00	6 seconds ago	106ms		25.808	1578.753
☐	run_2011-02-08 00:00:00 : 2011-02-14 23:00:00	6 seconds ago	105ms		16.835	604.725
☐	run_2011-01-29 00:00:00 : 2011-02-07 23:00:00	7 seconds ago	72ms		13.943	471.505
☐	Refrence_run	5 minutes ago	9.0s		9.436827956...	164.7628134...

☐ Show more columns (6)

Jun -1 - Jan 28